# JATIN SATYAM

**MS Robotics & Autonomous Systems @ ASU · Mechanical Design · Thermal & Structural Analysis**

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## SUMMARY

Mechanical engineer with cryogenic and defense hardware experience across three internships - cryocooler mechanism design in SolidWorks and NX, thermal validation testing, and ANSYS FEA/CFD - who took a gyro motor rotor from 8 CAD iterations through rotordynamic screening to instrumented FFT validation at 28,000 rpm. B.E. Mechanical Engineering (University of Mumbai); M.S. Robotics & Autonomous Systems at ASU; open to internships from Dec 2026 and full-time from May 2027.

## EXPERIENCE

**Research Assistant** Aug 2026 – Present

*Robotics Systems Instructional Lab, Arizona State University*

- Built and commissioned a second bench of a two-robot semiconductor wafer-handling cell (SCARA and six-axis arms, conveyor, Micro850 PLC), and turned both benches into labs for an ASU short-course manual
- Used the cell's NVIDIA Isaac Sim digital twin to find 2 interlock changes that cut the real cell's cycle time 26% (139.9 to 103.9 s, reproduced in 3 of 3 runs); its wafer contact physics matched the bench (2.6 mm slide predicted, 2.1-2.5 mm measured)

**Mechatronics Engineer Intern** Jan 2025 – Jul 2025

*Paras Defence & Space Technologies Ltd., Navi Mumbai, India · full-time at graduate-engineer-trainee scope*

- Designed and iterated cryocooler mechanisms in SolidWorks and NX - moving-assembly tolerances, interfaces, manufacturability - and led 25+ instrumented thermal runs on 3 assemblies, driving design changes that cut prototype rework 12%
- Built a vacuum-insulated LN2 generation unit sized to minimize parasitic heat leak, and ran a UAV structural lightweighting study achieving 8% mass reduction via material and topology changes
- Delivered 30+ CAD models and drawing revisions with design and fabrication teams while managing 3 project lifecycles in PLM - design-review turnaround down 20%, with no version drift between design and fabrication

**CFD Intern** Aug 2024 – Nov 2024

*MARS Exploration Pvt. Ltd (remote) · AS9100D-certified aerospace engineering firm*

- Performed 3D CFD analysis of NACA 23012 and NACA 4412 airfoils in ANSYS Fluent 2024 R2 (10-degree AoA, 48 m/s) - boundary-layer-refined meshing, solver setup, convergence, and separation/vortex post-processing

**R&D Engineer Intern** May 2023 – Aug 2023

*TriQual Cryogenics Pvt Ltd, Navi Mumbai, India*

- Contributed to mechanical design and vibration control for Stirling-cycle cryocoolers operating near 70 K, and supported gyro motor and helium compressor development under Make-in-India localization

## PROJECTS

**Gyro Motor Rotor Development** *· SolidWorks · ANSYS Rotordynamics · PHOTON+ FFT* Industry project · 2024

- Developed the rotor for a gyro motor used in gyro-stabilized defense platforms - 8 assembly iterations (straight and inclined bar, 8-11 mm) in EN24 and EN8, with component breakdowns and assembly drawings for manufacture
- Screened all 8 iterations in ANSYS - Campbell critical speeds, damped natural frequencies, stress versus yield at a 28,000 rpm / 466.67 Hz operating point - selecting the 9 mm straight-bar design on a 5,803 rpm critical-speed margin
- Manufactured and validated it on a physical test rig (accelerometer and tachometer, PHOTON+ analyzer with RT Pro FFT) - peak response 324.37 Hz, deviation from simulation attributed to unmodeled stator and housing elements · 2nd Prize, National Project Poster Competition, IEI-FCRIT 2024

**SipSafe Smart Cup - Assistive Device for Dysphagia Patients** *· Mechanical Design · Prototyping* ASU · 2026

- Owned the initial design and the complete mechanical design of a smart cup for dysphagia patients, built by a 5-person team with a Dignity Health clinical partner; co-inventor on an ASU invention disclosure

## SKILLS

**CAD & Mechanical Design:** SolidWorks, Siemens NX, CATIA V5, Fusion 360, AutoCAD, GD&T, DFM, Tolerance Analysis

**Simulation & Analysis:** ANSYS (Fluent, Modal/Rotordynamics), COMSOL, FEA, CFD, Thermal & Structural Analysis, ROS 2 Jazzy, Gazebo Harmonic, NVIDIA Isaac Sim

**Test & Instrumentation:** Test Rig Design, DAQ, LabVIEW, FFT Signal Analysis, Accelerometry, Design of Experiments

**Manufacturing, Systems & Tools:** PLM & Configuration Management, FMEA, Root-Cause Analysis, V&V, Prototype Development, MATLAB/Simulink, Python

## EDUCATION

**M.S. Robotics & Autonomous Systems (Systems Eng.), Arizona State University · GPA 3.67/4.0** Expected May 2027

**B.E. Mechanical Engineering with Major (Honors) in Electric Vehicles, University of Mumbai** 2020 – 2024